TEACHER GUIDE · GRADE 2 · BLOCK CODING

Read Both Scripts

Quick facilitation notes & answer key

Goal

Students read two concurrent scripts that now have TWO moves each, and find where each sprite lands. Reading and running code that involves concurrent events is the new Grade-2 idea in C3.1/C3.2. The two-move scripts give more code to trace while keeping the two scripts genuinely parallel.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: two counters (green for Bit, purple for Pixel) on a floor number path 0-5; say 'green flag!' and walk each through its two moves.

Run it (about 20 min)

1. I DO: read Bit's script ONLY first — 'start 0, forward 3 → 3, back 1 → 2.' Trace one move at a time.
2. WE DO: then read Pixel's script — 'start 5, back 2 → 3, forward 1 → 4.' Now press the green flag so BOTH run together.
3. YOU DO: Exercise 3 — students record both landings (Bit 2, Pixel 4) and notice they are different numbers.
4. Connect: 'Reading two scripts at once still means reading each one move by move.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 3 → 3 → back 1 → 2. Bit lands on 2.

Ex 2 — Pixel: 5 → back 2 → 3 → forward 1 → 4. Pixel lands on 4.

Ex 3 — when the green flag starts both at once, Bit lands on 2 and Pixel lands on 4. They are NOT on the same number, so they do not meet.

Interaction note: the two scripts run independently and at the same time — each sprite's moves change only its own position, and the green flag launches both together.

Watch for & differentiate

Common error: running one whole script, then the other, as if they were sequential — stress that the green flag starts BOTH at the same time (concurrent), even though we read them one move at a time.

Common error: dropping the second move — remind students that each script has TWO moves to read in order.

Simplify: trace Bit's two moves alone, then Pixel's two moves alone, before doing both together.

Extend: ask how many numbers apart the two sprites land (2 and 4 are two apart).

Success indicator: the child finds Bit on 2 and Pixel on 4 and explains that both scripts ran at the same time from the green flag.